

## 184.702 Machine Learning – Exercise 3.2 Deep Learning for Image or Text Tasks

*Note: please first read the general overview document for the 3<sup>rd</sup> exercise!*

### Topic 3.2: Deep Learning

*N.b.: if you want to work with Deep Learning, but on more advanced topics, also take a look at the task on security (adversarial machine learning: inversion, evasion, poisoning attacks) and on synthetic image generation (using generative adversarial networks) in exercise Topic 3.1.*

**N.b. #2:** You can reuse existing materials / tutorials / etc. from the Internet, but (i) you need to clearly quote / cite what you use, (ii) there still needs to be an original contribution from your side.

#### 3.2.1: Image **OR** Text classification - Feature Extraction & Shallow vs. Deep Learning

The main goal of this exercise is to get a feeling and understanding on the importance of representation and extraction of information from complex media content, in this case images or text (just one of them, not both!). You will work with a datasets that has an image/text classification target.

Main difference between convolution (CNN) and traditional image processing:

Both uses filter kernels (matrices) and apply it on top and get a result. CNN however learns how these kernels need to look like, while the image processing has some standard predefined ones (identity matrix > weirdly just a one in the middle not along the axis, ...)

(1) In the first step, you shall try to find a good classifier with „traditional“ methods. Thus, pick

For Images

- One simple feature representation (such as a colour histogram, see also the examples provided in TUWEL; you need to use only one such (the most powerful) representation, e.g. the 3-channel histograms), **and**
- A feature extractor based on key point detection or similar, such as SIFT ([https://en.wikipedia.org/wiki/Scale-invariant\\_feature\\_transform](https://en.wikipedia.org/wiki/Scale-invariant_feature_transform)) and subsequent visual bag of words (e.g. <https://kushalvyas.github.io/BOV.html> for python), or a similar powerful approach (such as SURF, ORB, ..).

For Text

- One feature extractor based on e.g. Bag Of Words, or n-grams, or similar.

You shall evaluate these features on two shallow algorithms, optimising the parameter settings to see what performance you can achieve, to have a baseline for the subsequent steps.

(2) Then, try to use deep learning approaches, such as **convolutional neural networks (for images, but also potentially for text, e.g. with 1D convolutions)**, recurrent neural networks (primarily for

Difference between CNN and MLP (multi-layer perceptron):

MLP > Each input and each weight is a scalar value. Each Output is the sum of the inputs multiplied by the weights

CNN > Each input is a tensor (photo with its pixels basically), each weight (kernel!) is a tensor. And then each output is the sum if the inputs that are convolved by the weights

**text), combinations thereof, or other approaches** (but not just simple MLPs); compare the performance to the shallow learning approaches. **Try at least two different DL architectures.** They can be existing, well-known ones that you either re-use or adapt.

For text analysis, you should also test whether employing word (or character) embedding layers, e.g. using word2vec, GloVe, etc., changes the results. You can re-use pretrained embeddings.

Compare not just the overall measures, but perform a detailed comparison and analysis per class (confusion matrix), to identify if the two approaches lead to different types of errors in the different classes, and also try to identify other patterns.

For images, you can base your DL implementation on the tutorial provided by colleagues at the institute, available at

[https://github.com/tuwien-musicir/DL\\_Tutorial/blob/master/Car\\_recognition.ipynb](https://github.com/tuwien-musicir/DL_Tutorial/blob/master/Car_recognition.ipynb) (you can also check the rest of the repository for interesting code; credit to Thomas Lidy (<http://www.ifs.tuwien.ac.at/~lidy/>)). Mind also that you will find plenty of examples on how to create and train CNNs / RNNs in various frameworks - tensorflow, keras, pytorch, ....

Also perform a detailed comparison of runtime, considering both time for training and testing, including also the feature extraction components.

For the **datasets** you shall work with, pick two text/image datasets, from the list of suggestions below. If you have proposals for other datasets, please inform me ([mayer@ifs.tuwien.ac.at](mailto:mayer@ifs.tuwien.ac.at)), and we can see if the dataset is fit.

- For Images :
  - The German Traffic Sign Recognition Benchmark (GTSRB), <http://benchmark.ini.rub.de/>
  - PubFig83: <http://vision.seas.harvard.edu/pubfig83/>  
Text
  - CIFAR-10: <https://www.cs.toronto.edu/~kriz/cifar.html>
  - Tiny ImageNet: <https://tiny-imagenet.herokuapp.com/>
  - FashionMNIST: <https://github.com/zalandoresearch/fashion-mnist>
  - Labeled Faces in the Wild, <http://vis-www.cs.umass.edu/lfw/>, for creating a classifier recognising different people (1 person == 1 class). See also <https://scikit-learn.org/0.19/datasets/#the-labeled-faces-in-the-wild-face-recognition-dataset>. For this dataset, it is best to use a subset only, of people that have a reasonable number of images (e.g. >=20).
  - CelebA: <https://plg.uwaterloo.ca/cgi-bin/cgiwrap/gvcormac/foo07>. Use this dataset, as above the Labeled Faces in the Wild, for a face recognition dataset; take a similar subset.
- For Text Data:
  - 20 newsgroups: <http://qwone.com/~jason/20Newsgroups/>
  - Reuters: <http://www.daviddlewis.com/resources/testcollections/reuters21578/>

- Fake News Dataset: [https://github.com/GeorgeMcIntire/fake\\_real\\_news\\_dataset](https://github.com/GeorgeMcIntire/fake_real_news_dataset)
- Anything else :-)

Recommendations for CNNs specifically, but also for RNNs:

- Use architectures of your choice – you can work with something simple like a LeNet, optimised for smaller network structures like SqueezeNet, or more advanced architectures (where maybe transfer learning is required for efficiency reasons, see below).
- Use as well data augmentation (you can reuse the code from the tutorial), and compare it to the non-augmented results
- Also use transfer learning of pre-trained models
- As stated above, mind that there is plenty of example code available that you can reuse (in your report, state what sources you have utilised).
- The main focus is on analysing the differences in traditional vs. deep learning.

So, what does a CNN do?

- Reduces the number of inputs (e.g. pixels)
- By convolution, but mostly subsampling (pooling)
- Tolerates shifts in the location
- By not directly learning a mapping from pixel position to output
- Uses correlations of pixel patterns
- Considers regions of image, not single pixels
- Each cell in feature map corresponds to group of neighbouring pixels
- Uses the same filters in all regions

Often used: ReLU for the activation function because of higher gradients also for larger inputs > Avoids vanishing gradients. Also no saturation, low comp.cost

Which size for kernels? (“receptive fields”)  
 3x3 kernel: Can only see small structure  
 7x7 kernel: Sees more, but more weights  
 224x224 kernel: Possibly sees full input  
 Equivalent to fully-connected then.

Onto Adam (Adaptive Moment Estimation): Optimization methods that adapt the learning rate for each parameter

- Compute velocity (first moment): exponential moving average over past gradients (as before)
- Compute second moment estimate: exponential moving average over past gradient magnitudes
- Move according to velocity, divided by second moment
- Intuition: counter notoriously small gradients by upscaling, and large gradients by downscaling
- Separately for each weight

### 3.2.2: Image-to-image processing via Deep Learning (e.g. CNN-based Image upscaling)

This task aims to make you familiar with some image-to-image deep learning tasks. More precisely, you can work on (a) image upscaling, (b) image deblurring, or (c) image colorization (pick only one!). You shall on purpose distort your images, and then restore them as close as possible to the original state.

We expect you to perform the following steps:

1. Pick a CNN that is applicable for your image-to-image task. For instance, you can use the following approaches. *Please note that we did not test the mentioned implementations/github repositories, so issues are not excluded. Feel free to use any other relevant solution you find.*

a) Super-Resolution Convolutional Neural Network (SRCNN)

- Original paper: <https://arxiv.org/abs/1501.00092>
- (Unofficial) Pytorch implementation: <https://github.com/yjn870/SRCNN-pytorch>

b) DeepDeblur

- Original paper: [https://openaccess.thecvf.com/content\\_cvpr\\_2017/papers/Nah\\_Deep\\_Multi-Scale\\_Convolutional\\_CVPR\\_2017\\_paper.pdf](https://openaccess.thecvf.com/content_cvpr_2017/papers/Nah_Deep_Multi-Scale_Convolutional_CVPR_2017_paper.pdf)
- Pytorch implementation: <https://github.com/SeungjunNah/DeepDeblur-PyTorch>

c) Colorful Image Colorization

- Original paper: <https://arxiv.org/abs/1603.08511>
- Official Pytorch implementation (testing only): <https://github.com/richzhang/colorization>
- (Unofficial) Pytorch implementation: <https://github.com/Time0o/colorful-colorization>
- (Unofficial) Tensorflow (Keras) implementation: <https://github.com/RoddeTheR/deep-learning-image-colorization>

2. Pick a dataset (different from the implementations above) for your task. Keep in mind that you can use any high-resolution dataset (for instance, COCO <https://cocodataset.org/#home>; subsets of ImageNet, ...) and transform it for your task (downscaling, adding blur, converting to grayscale).

3. Train a network (or fine-tune a pre-trained one) on your dataset.

4. Evaluate the performance of the model

- Qualitatively by visualizing and analysing some results (e.g. compare original versus restored (upscaled, ...) image)
- Quantitatively, e.g. by computing some image comparison metric (e.g. the PSNR score if it is applicable for your task, or L1 / L2 distances, ...), or by comparing the performance of your original vs. the restored (upscaled, ...) images when used as a training set e.g. on a classification task.

### 3.2.3: Next-word prediction (Language Modelling) using Deep Learning

Next-word prediction is a task where, given an incomplete sequence of words, the most likely next word is predicted. This is useful for several application scenarios, e.g. auto-completion in a search engine, or especially predicting likely next words on devices with limited input means, such as mobile phones. Recently, deep learning has been utilised extensively for this task.

In this exercise, you shall evaluate one deep learning approach for next word prediction, on any of the text corpora mentioned above; basically, you shall use the corpus to use the existing sequences as training data, and keep a holdout dataset of not used sentences to test the completion. One difficulty in this task in general is evaluation, as text is ambiguous, and multiple words would be acceptable as “correct” next word. Thus, also consider how to evaluate your model; automated methods are fine, but do not restrict yourself for accuracy only; while this is a good baseline, it does not capture the above-mentioned also acceptable synonyms, so also consider advanced methods to measure correct predictions; you can consider different metrics, and/or consider using embeddings to consider semantic similarity, .... Also consider a small manual / subjective assessment.

Sources: “A Survey on Neural Network Language Models”, <https://arxiv.org/abs/1906.03591>